Technical specification template: Low-Voltage Switchgear and Assemblies

OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. Package revision A6. Generated 2026-11-08 20:00 UTC. Package version 0.1.0.dev0.

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Document control

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1. Scope

This template defines the technical requirements for low-voltage switchgear and assemblies for the OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. It is to be completed and issued by the project. Items in [square brackets] are decisions or data the project must supply. Preliminary values from the Ohmetry reference configuration are shown with their data status and must be replaced or confirmed.

2. Applicable standards

- IEC 61439-1 and IEC 61439-2 (low-voltage switchgear and controlgear assemblies), design verification
- IEC 60947-2 (circuit-breakers)
- IEC 61869 series (instrument transformers)
- Local regulations [project to add]
- Editions and applicability of every standard above are NOT VERIFIED. The project shall confirm the current editions and any national deviations.

3. Service conditions and system data

Item	Requirement or preliminary value	Data status
Nominal LV system	400 V line-to-line, 60 Hz, 3-phase	PRELIMINARY_ASSUMPTION. Utility confirmation is the owner's statement, not independently verified
MV supply	13.8 kV	PRELIMINARY_ASSUMPTION. To be confirmed with the utility
Design ambient temperature	30 degC	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Site altitude	0 m	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Prospective fault level	[To be calculated: short-circuit	NOT CALCULATED

	study pending utility, transformer and generator data]	

4. Equipment ratings

Ratings below are preliminary values from the master configuration unless marked otherwise. They are not manufacturer-confirmed and are not an equipment selection.

Item	Requirement or preliminary value	Data status and note
Rated voltage and frequency	400 V, 60 Hz	PRELIMINARY_ASSUMPTION
Busbar rated current	[At least the incomer design current after derating; preliminary 4,136 A (path-out)]	Preliminary requirement, pending manufacturer data
Rated short-time withstand (kA, 1 s) and peak	[To be at least the calculated LV fault level]	NOT CALCULATED: utility, transformer impedance and generator data pending
Form of separation, IP rating	[Project to select]	Project decision
Circuit-breaker breaking and making capacity	[At least the fault level at each location]	Pending manufacturer data

5. Technical requirements

5.1 The assembly shall be design verified per IEC 61439-1 and -2; the verification evidence (test, calculation or design rule) shall be supplied for each rating, with the assembly manufacturer identified.

5.2 The short-circuit withstand shall exceed the prospective fault level including the peak, at the busbars and at each outgoing device.

5.3 Circuit-breakers shall have breaking capacity above the fault level at their location; time-current characteristics shall be supplied for the coordination study.

5.4 Temperature rise at the rated current and the site ambient shall be evidenced; derating of busbars at altitude and ambient shall be stated.

5.5 Metering, monitoring, interlocking, transfer scheme and arc-fault mitigation: [project to define].

6. Vendor data schedule

The tenderer shall complete the schedule below with its offer: the value offered, its unit, and the name and revision of the document that states it. A value without a source document is not accepted. The parameters are those in the manufacturer-data register; supplied values are imported into the calculations through the data override mechanism, and a value stays NOT VERIFIED until a reviewer records an attestation against its source (name, organisation, date, evidence reference, scope).

Parameter id	Parameter	Datasheet required	Current status	Value offered	Unit	Document and revision
distribution.LV-MSB-A.rated_current_a	LV-MSB-A: rated_current_a	LV switchgear data sheet and IEC	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]

		61439 verification evidence				
distribution.LV-MSB-A.short_circuit_rating_ka	LV-MSB-A: short_circuit_rating_ka	LV switchgear data sheet and IEC 61439 verification evidence	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
distribution.LV-MSB-A.form_of_separation	LV-MSB-A: form_of_separation	LV switchgear data sheet and IEC 61439 verification evidence	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
distribution.LV-MSB-A.ip_rating	LV-MSB-A: ip_rating	LV switchgear data sheet and IEC 61439 verification evidence	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
distribution.LV-MSB-A.verification_evidence	LV-MSB-A: verification_evidence	LV switchgear data sheet and IEC 61439 verification evidence	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
distribution.ACB-TX-A.breaking_capacity_ka	ACB-TX-A: breaking_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
distribution.ACB-TX-A.making_capacity_ka	ACB-TX-A: making_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
distribution.ACB-TX-A.time_current_curve_ref	ACB-TX-A: time_current_curve_ref	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
distribution.ACB-TX-A.settings	ACB-TX-A: settings	Circuit breaker or relay data sheet with time-current curves and test	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]

		certificates				
distribution.ACB-GEN-A.breaking_capacity_ka	ACB-GEN-A: breaking_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
distribution.ACB-GEN-A.making_capacity_ka	ACB-GEN-A: making_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
distribution.ACB-GEN-A.time_current_curve_ref	ACB-GEN-A: time_current_curve_ref	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
distribution.ACB-GEN-A.settings	ACB-GEN-A: settings	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
distribution.ACB-UPS-IN-A.breaking_capacity_ka	ACB-UPS-IN-A: breaking_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
distribution.ACB-UPS-IN-A.making_capacity_ka	ACB-UPS-IN-A: making_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
distribution.ACB-UPS-IN-A.time_current_curve_ref	ACB-UPS-IN-A: time_current_curve_ref	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
distribution.ACB-UPS-IN-A.settings	ACB-UPS-IN-A: settings	Circuit breaker or relay data sheet with time-current	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]

		curves and test certificates				

Model, manufacturer, country of manufacture, and the type-test reports for the offered design: [vendor to complete].

7. Compliance schedule

The tenderer shall state, for every standard in section 2 and every technical requirement in section 5, whether the offer complies. 'Complies' needs an evidence reference (type-test report, certificate, calculation, drawing or data sheet with revision). A statement without evidence is recorded as not demonstrated. The requirement references below are the numbered items of section 5 of this template; no clause of any standard is quoted.

Standard named in section 2	Edition offered	Complies / complies with comment / does not comply	Evidence reference	Comment or deviation
IEC 61439-1 and IEC 61439-2 (low-voltage switchgear and controlgear assemblies), design verification	[vendor to complete]	[vendor to complete]	[vendor to complete]	[vendor to complete]
IEC 60947-2 (circuit-breakers)	[vendor to complete]	[vendor to complete]	[vendor to complete]	[vendor to complete]
IEC 61869 series (instrument transformers)	[vendor to complete]	[vendor to complete]	[vendor to complete]	[vendor to complete]
Local requirements [project to add]	[vendor to complete]	[vendor to complete]	[vendor to complete]	[vendor to complete]

Requirement (section 5)	Complies / complies with comment / does not comply	Evidence reference	Comment or deviation
5.1	[vendor to complete]	[vendor to complete]	[vendor to complete]
5.2	[vendor to complete]	[vendor to complete]	[vendor to complete]
5.3	[vendor to complete]	[vendor to complete]	[vendor to complete]
5.4	[vendor to complete]	[vendor to complete]	[vendor to complete]
5.5	[vendor to complete]	[vendor to complete]	[vendor to complete]

8. Inspection and quality requirements

8.1 Review of design verification evidence, drawings and the ITP.

8.2 Inspection of busbar supports, clearances, wiring and labelling.

8.3 Routine tests on every unit as required by the applicable standard and this specification.

8.4 Type-test reports or evidence for the design, issued by an accredited or recognised laboratory [project to define acceptable evidence].

8.5 Inspection and test plan (ITP) approved before manufacture, naming witness and hold points.

8.6 Calibration certificates for test instruments.

9. Factory acceptance test (FAT) requirements

9.1 Routine verification per IEC 61439: wiring, operation, dielectric, protective circuit continuity, mechanical function.

9.2 Breaker functional tests and settings checks; transfer scheme simulation.

9.3 Witnessing: [owner / independent inspector to attend].

9.4 A FAT procedure shall be submitted for approval before the test; the results shall be issued as a signed report, with deviations and their disposition recorded.

9.5 Acceptance criteria are taken from the applicable standard and this specification [project to define]; a test passed against a criterion not stated in advance is not accepted.

9.6 Equipment shall not be shipped until the FAT report is accepted or a concession is recorded and approved by the owner.

10. Site acceptance test (SAT) requirements

10.1 Insulation resistance; mechanical operation and interlocking (including the utility-to-generator interlock); protection settings as set by the approved study; metering and communications checks.

10.2 A SAT procedure shall be submitted for approval before the tests. It states the test, the method, the instrument, the acceptance criterion and the witness, with the criteria taken from the applicable standard and this specification [project to define].

10.3 SAT is carried out on the installed equipment with its final connections, after the installation inspection and before commissioning loads are applied.

10.4 Results are recorded on signed sheets. Any deviation, retest and concession is listed with its disposition. A SAT is not passed while a deviation is open.

10.5 SAT results are compared with the FAT results; a significant difference is investigated and recorded.

11. Commissioning requirements

11.1 Torque and insulation resistance, phase sequence, protection tests, interlock and transfer tests.

11.2 Load and thermography checks [project to define].

11.3 Commissioning records shall name the test, the instrument and its calibration, the date and the person, and shall be retained with the verification register.

11.4 Integrated systems testing of the whole power chain (utility loss, generator start and transfer, UPS on battery, loss of one power path, return to normal) is carried out under a project procedure [project to define]. This equipment's commissioning is not complete until it has taken part in that test.

12. Submittals and documentation

Document	When required	Status
Completed vendor data schedule (section 6) and compliance schedule (section 7)	With the tender	[vendor to complete]
Datasheets and general arrangement drawings	With the tender, final before manufacture	[vendor to complete]
Type-test reports and certificates for the offered design	With the tender	[vendor to complete]
Inspection and test plan (ITP)	Before manufacture	[vendor to complete]
FAT procedure and report	Before and after the FAT	[vendor to complete]
Installation instructions and SAT procedure	Before delivery	[vendor to complete]
Operation and maintenance manuals, as-built drawings and final data	Before commissioning; as-built at handover	[vendor to complete]
Recommended spare parts and special tools lists	With the tender and at handover	[vendor to complete]
Training plan and materials [project to define]	Before commissioning	[vendor to complete]
Warranty terms [project to define]	With the tender	[vendor to complete]

13. Open items (NOT VERIFIED)

- All ratings in section 4 marked PRELIMINARY or UNVERIFIED are open until confirmed from manufacturer data and the project studies.
- The prospective fault level is NOT CALCULATED until the utility, transformer and generator data are supplied.
- Standards editions and local requirements are NOT VERIFIED.
- The vendor data schedule (section 6) and the compliance schedule (section 7) are blank until a tenderer completes them.
- FAT, SAT and commissioning acceptance criteria are [project to define].
- This template has not been reviewed by an independent engineer.